

Linyi (Tiger) Hou

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EDUCATION

Ph.D. in Aerospace Engineering, University of Illinois Urbana-Champaign	Jun 2021 - May 2025
B.S. in AE, Minor in CompE, University of Illinois Urbana-Champaign	Aug 2017 - May 2021

SKILLS

Technical	celestial navigation, astrodynamics, modeling & simulation, data analysis
Collaboration	technical/proposal writing, MS Office suite, L ^A T _E X, Git, GitHub, Trello
Programming	proficient: Python, C/C++, MATLAB beginner: Simulink, Julia, Javascript

EXPERIENCE

Graduate Researcher, University of Illinois Urbana-Champaign	Jun 2021 - present
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- ❖ Simulated photon time-of-arrival measurements taken by spacecraft navigation sensors.
- ❖ Wrote simulation software to facilitate the research and development of several novel spacecraft navigation and attitude determination techniques.
- ❖ Presented research results at conferences and in peer-reviewed journal publications.

Teaching Assistant, University of Illinois Urbana-Champaign	Jun 2021 - May 2024
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- ❖ Worked with 2-5 faculty and staff to teach lessons, hold office hours, and improve curriculum for courses in flight mechanics, systems engineering, and UAV controls.
- ❖ Assisted in developing code and instructions for a UAV lab based on C and Python.
- ❖ Created a UAV flight simulation to predict the performance of various PID controllers.

Software Engineering Intern, Collins Aerospace	May 2019 - Aug 2019
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- ❖ Migrated an avionics data interpreter from 16-bit to 32-bit using C++ and Python.
- ❖ Automated weekly builds and integration in TortoiseSVN using Python.

Team Lead/Advisor, NASA's RASC-AL Competition	2019, 2022 - 2023
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- ❖ Led a team of 10+ students to develop a reusable crewed lunar lander concept (structure, power, thermal, life support, orbital mechanics, budget, etc.).
- ❖ Advised two student teams who developed requirements and performed trade studies for lunar and Martian in-situ resource utilization architectures.

HONORS AND AWARDS

Aerospace Engineering Alumni Advisory Board Fellowship, UIUC	2024
Best Student Paper, 46 th AAS GN&C Conference, American Astronautical Society	2024
Yee Memorial Fellowship, UIUC	2021, 2024
David and Catherine Thompson Space Technology Scholarship, AIAA	2020